

REQUIREMENT ANALYSIS

ONE8 Footwear E-Commerce Database

1. Project Introduction

The **ONE8 Footwear E-Commerce Database** is designed to provide a secure and efficient database system for an online footwear shopping platform. The database stores and manages information related to customers, products, categories, brands, orders, payments, inventory, and administrators.

The system enables customers to browse footwear collections, search for products, add items to their shopping cart or wishlist, place orders, make secure online payments, and track deliveries. Administrators can efficiently manage products, inventory, customer details, and sales records.

The primary objective of this project is to develop a well-structured relational database that ensures efficient data storage, quick retrieval, and data integrity for an online footwear business.

2. Requirement Analysis

Requirement Analysis is the process of identifying, collecting, analyzing, and documenting the functional and non-functional requirements of a system before development begins.

For the **ONE8 Footwear E-Commerce Database**, the requirement analysis focuses on customer management, product management, order processing, payment handling, inventory management, and maintaining data integrity.

The requirements are divided into:

- Functional Requirements
 - Non-Functional Requirements
 - Software Requirements
 - Hardware Requirements
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3. Objectives

The main objectives of this project are:

- To design a normalized relational database.
 - To provide secure customer registration and login.
 - To manage footwear products efficiently.
 - To support shopping cart and wishlist functionality.
 - To process customer orders and payments securely.
 - To maintain inventory records accurately.
 - To generate sales and order reports.
 - To ensure data consistency and integrity.
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4. Scope

The scope of the ONE8 Footwear E-Commerce Database includes:

- Customer Registration
- User Login
- Product Management
- Category Management
- Brand Management
- Product Search
- Shopping Cart
- Wishlist
- Order Management
- Payment Management
- Delivery Tracking
- Inventory Management

- Product Reviews
- Admin Dashboard
- Sales Reports

Future Enhancements

- AI-based Product Recommendations
 - Personalized Shoe Size Suggestions
 - Mobile Application
 - Loyalty Rewards Program
 - Return and Refund Management
 - Analytics Dashboard
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5. Existing System

In the existing system, customers usually visit physical stores to purchase footwear. Product availability depends on store inventory, and billing is often handled manually. Customers have limited options to compare products and may spend significant time searching for suitable footwear.

Problems of Existing System

- Time-consuming shopping process.
 - Limited product availability.
 - Manual billing process.
 - Difficulty comparing products.
 - No online order tracking.
 - Limited shopping hours.
 - Travel expenses.
 - Poor inventory management.
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6. Proposed System

The proposed **ONE8 Footwear E-Commerce Database** provides an online platform where customers can purchase footwear anytime and from anywhere.

Features

- Customer Registration
- Secure Login
- Product Search
- Category & Brand Management
- Shopping Cart
- Wishlist
- Secure Online Payment
- Order Tracking
- Product Reviews
- Inventory Management
- Sales Report Generation
- Admin Dashboard

Benefits

- Saves customer time
 - Easy online shopping
 - Secure transactions
 - Faster order processing
 - Better inventory control
 - Improved customer satisfaction
 - Efficient business management
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7. Functional Requirements

Functional requirements describe the operations performed by the system.

Customer Module

- Customer Registration
- Login
- Profile Management
- Password Update

Product Module

- Add Products
- Update Products
- Delete Products
- View Product Details

Category Module

- Add Category
- Update Category
- Delete Category

Brand Module

- Add Brand
- Update Brand
- Delete Brand

Shopping Cart Module

- Add to Cart
- Remove from Cart
- Update Quantity

- View Cart

Wishlist Module

- Add Wishlist
- Remove Wishlist

Order Module

- Place Order
- Cancel Order
- View Order History
- Track Orders

Payment Module

- Online Payment
- Payment Confirmation
- Payment History

Admin Module

- Manage Customers
- Manage Products
- Manage Orders
- Manage Inventory
- Generate Reports

8. Non-Functional Requirements

Performance

- Fast query execution.
- Quick response time.

- Efficient database performance.

Security

- Secure user authentication.
- Encrypted passwords.
- Protected payment information.

Reliability

- Accurate data storage.
- Backup and recovery support.
- Continuous availability.

Usability

- Easy-to-understand database structure.
- Simple data management.

Scalability

- Support increasing customers and products.
- Easy expansion of database tables.

Maintainability

- Easy updates.
- Easy modification.
- Future feature support.

9. Software Requirements

Software

Purpose

Windows 10/11 Operating System

Software	Purpose
MySQL	Database Management System
MySQL Workbench	Database Design
SQL	Database Query Language
Visual Studio Code	Code Editor
Git	Version Control
GitHub	Repository Management
Microsoft Word	Documentation

10. Hardware Requirements

Hardware Specification

Processor	Intel Core i3 or above
RAM	4 GB Minimum
Storage	50 GB Free Space
Monitor	1366 × 768 Resolution
Keyboard	Standard Keyboard
Mouse	Optical Mouse
Internet	Broadband Connection

11. Modules

Module 1 – Customer Management

- Registration

- Login
- Profile Management

Module 2 – Product Management

- Add Product
- Update Product
- Delete Product
- View Product

Module 3 – Category & Brand Management

- Manage Categories
- Manage Brands

Module 4 – Shopping Cart

- Add Items
- Remove Items
- Update Quantity

Module 5 – Wishlist

- Add Wishlist
- Remove Wishlist

Module 6 – Payment

- Payment Processing
- Payment Status

Module 7 – Order Management

- Place Order
- Order Tracking
- Order History

Module 8 – Inventory Management

- Stock Management
- Inventory Updates

Module 9 – Admin Dashboard

- Customer Management
 - Product Management
 - Order Management
 - Sales Reports
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12. Conclusion

The **ONE8 Footwear E-Commerce Database** provides a secure, efficient, and well-structured database solution for managing an online footwear business. It simplifies customer management, product handling, order processing, payment management, and inventory control while ensuring data integrity through relational database concepts. The system supports future enhancements such as AI-based recommendations, mobile applications, and advanced analytics, making it a scalable solution for modern e-commerce platforms.